

DealFlow360 — Application Flow (Admin & Customer)

ADMIN FLOW

Step 1 — Login

Open `http://localhost:5173/login`
→ Click "Admin" demo card (pre-fills `admin@dealflow.com / Admin@123`)
→ Click "Sign In"
→ POST `/api/auth/login`
→ JWT access token stored in Zustand + localStorage
→ Refresh token stored as httpOnly cookie
→ Redirected to `/dashboard`

Step 2 — Dashboard

`/dashboard`
■■■ KPI Cards (fetched from GET `/api/dashboard/metrics`)
■ ■■■ Total Quotations
■ ■■■ Confirmed Deals & Value
■ ■■■ Pending Approvals
■ ■■■ Active Subscriptions
■ ■■■ Avg Deal Size
■
■■■ Charts
■ ■■■ Revenue Trend (AreaChart – Recharts)
■ ■■■ Pipeline by Status (BarChart – Recharts)
■
■■■ Anomaly Alerts
■ ■■■ Stalled Deals (no update in 7 days)
■ ■■■ Discount Anomalies (unusually high discount %)
■ ■■■ Expiring Quotations (validity < 7 days)
■
■■■ Socket.io joins `'dashboard'` room
→ auto-refreshes on `'dashboard:update'` event

Step 3 — Product Catalog Setup

`/products` (ADMIN only)
■■■ View all products (GET `/api/products`)
■■■ Add New Product
■ ■■■ Fill: Name, SKU, Base Price, Cost Price, Tax %, Category
■ ■■■ Upload product image (FastAPI UploadFile → `uploads/products/`)
■ ■■■ POST `/api/products` → product created
■
■■■ Edit Product → PUT `/api/products/:id`
■■■ Add Variants (size/color) → POST `/api/products/:id/variants`
■■■ Delete Product → DELETE `/api/products/:id`

Step 4 — Configure Discount Tiers

`/discount-tiers` (ADMIN, SALES_MANAGER)
■■■ View 3 tiers: BRONZE, SILVER, GOLD
■■■ Set max discount % per tier
■ ■■■ BRONZE → 5% (most basic customers)
■ ■■■ SILVER → 10%
■ ■■■ GOLD → 20% (enterprise customers)
■■■ PUT `/api/products/discount-tiers/:tier` → saved

Step 5 — Set Up Upsell Rules

`/upsell-rules` (ADMIN, SALES_MANAGER)
■■■ Create rule: IF product A in cart → SUGGEST product B
■■■ Trigger types:
■ ■■■ PRODUCT_COMBO – specific product pairing
■ ■■■ QUANTITY_THRESHOLD – qty > N triggers suggestion
■ ■■■ CUSTOMER_TIER – only show upsell to GOLD customers
■■■ POST `/api/products/upsell-rules` → rule saved
(QuotationBuilder auto-shows these suggestions)

Step 6 — Manage Warehouses & Stock

```
/warehouses (ADMIN, SALES_MANAGER, FINANCE)
■■■ View 3 warehouses (Main, East, West)
■■■ Stock grid: rows = products, columns = warehouses
■■■ Adjust stock → PUT /api/fulfillment/warehouses/:wId/stock/:pId
■■■ Low stock warning (< 10 units → red badge)
```

Step 7 — Manage Users

```
/users (ADMIN only)
■■■ View all users (GET /api/auth/users)
■■■ Create new user
■    ■■■ Assign role: ADMIN / SALES_REP / SALES_MANAGER / FINANCE / CUSTOMER
■    ■■■ POST /api/auth/users
■■■ Deactivate user → PUT /api/auth/users/:id/status
■■■ Reset password → PUT /api/auth/users/:id/reset-password
```

Step 8 — Create a Quotation

```
/quotations/new (ADMIN, SALES_REP, SALES_MANAGER)
■
■■■ Fill Header
■    ■■■ Customer name
■    ■■■ Validity date
■    ■■■ Payment terms
■
■■■ Add Line Items
■    ■■■ Click "Add Product" → ProductPicker modal opens
■    ■■■ Search by name or SKU
■    ■■■ Select product → line added with base price & cost
■    ■■■ Adjust: Quantity, Discount %, Tax %
■    ■■■ LiveMarginBar updates in real-time:
■        ■■■ LIVE TOTAL (incl. tax)
■        ■■■ GROSS MARGIN % (with color bar: red/yellow/green)
■        ■■■ RISK SCORE (0-15, glowing LED)
■        ■■■ APPROVAL PATH hint
■
■■■ Upsell Suggestions appear automatically
■    ■■■ (based on UpsellRules configured in Step 5)
■
■■■ Save Draft → POST /api/quotations (status: DRAFT)
■
■■■ Submit for Approval → PUT /api/quotations/:id/submit
    → status becomes PENDING_MANAGER
    → Notification sent to Sales Manager
    → Socket.io emits <approval:new>; to <approvers>; room
```

Step 9 — Approve Quotations

```
/approvals (SALES_MANAGER, FINANCE, ADMIN)
■
■■■ Stage 1: PENDING_MANAGER queue
■    ■■■ View all quotations needing manager approval
■    ■■■ Click quotation → see line items, discount, risk score
■    ■■■ Approve → PUT /api/quotations/:id/decision { action: <approve>; }
■        → If discount > 10% → status: PENDING_FINANCE
■        → If discount ≤ 10% → status: APPROVED
■    ■■■ Reject → status: REJECTED, comment saved
■
■■■ Stage 2: PENDING_FINANCE queue (Finance user)
■    ■■■ Same flow – Finance reviews high-discount deals
■    ■■■ Approve → status: APPROVED
■
■■■ Batch Actions
■    ■■■ Select multiple quotations via checkboxes
■    ■■■ "Approve All" / "Reject All" → POST /api/quotations/batch-decision
```

Step 10 — Send to Customer

/quotations/:id (after APPROVED status)

- - Click "Send to Customer";
 - → PUT /api/quotations/:id/send
 - → Backend generates portal_token (UUID) on quotation
 - → Email sent to customer with portal URL
 - → status: SENT_TO_CUSTOMER
 -
 - Portal URL format:
http://localhost:5173/portal/{portal_token}
-

Step 11 — Monitor Negotiation

After customer opens portal and counter-offers:

- - Admin/Rep sees notification (bell icon in Navbar)
 - Socket.io: 'negotiation:message' arrives in quotation room
 - View negotiation messages on quotation detail page
 - Customer confirms → status: CONFIRMED
-

Step 12 — Fulfillment

/fulfillment (after CONFIRMED)

- - Quotation appears in fulfillment list
 - View split plan (if stock spread across warehouses)
 - ■■■ GET /api/fulfillment/:id/split
 - ■■■ Shows: Warehouse A → 3 units, Warehouse B → 2 units
 - Accept Split → POST /api/fulfillment/:id/accept-split
 - Stock deducted from warehouse automatically
-

Step 13 — Generate Invoice

/invoices

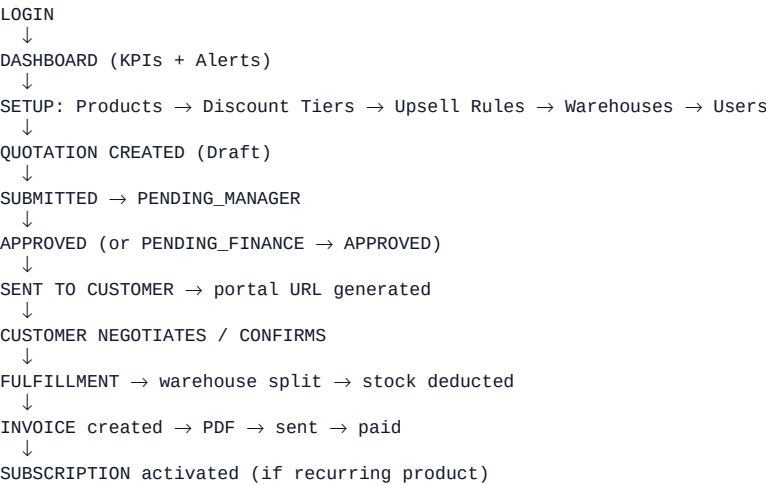
- - Create Invoice from confirmed quotation
 - → POST /api/invoices
 - Download PDF → GET /api/invoices/:id/pdf
 - (Axios: responseType: 'blob' → browser download)
 - Send Invoice → PUT /api/invoices/:id/send (email)
 - Mark as Paid → PUT /api/invoices/:id/pay
-

Step 14 — Subscription (for SaaS products)

/subscriptions

- - Create subscription from confirmed quotation
 - → POST /api/subscriptions/:quotationId
 - View: plan, billing cycle, MRR, renewal date
 - Cancel → PUT /api/subscriptions/:id/cancel
-

Admin Full Lifecycle Summary



CUSTOMER FLOW

Customers do NOT log into the main app. They access a **magic link URL** sent by email. No username/password needed — the token IS the credential.

Step 1 — Receive Email

Customer receives email:
Subject: "Your Quotation QT-2024-001 from DealFlow360"
Body: "Click here to view and respond:
http://localhost:5173/portal/portal-token-acme-004"

(OR scan QR code shown in the email / on the portal)

Step 2 — Open Customer Portal

http://localhost:5173/portal/{token} ← NO login required

→ React reads token from URL: useParams()
→ API call: GET /api/quotations/portal/{token}
→ Backend validates token → returns quotation data
→ CustomerPortal.jsx renders (944 lines)

Step 3 — View Quotation Details

Customer sees:

- Company branding header
- Quotation number, issue date, validity date
- Expiry countdown ("Expires in 5 days")
- Status badge (SENT_TO_CUSTOMER / UNDER_NEGOTIATION / CONFIRMED)
- Line Items Table
 - Product name, description, SKU
 - Quantity × Unit Price
 - Discount applied
 - Tax (GST 18%)
 - Line total
- Summary
 - Subtotal
 - Tax amount
 - TOTAL (bold, large)
- Payment Terms & Notes
- QR Code
 - Scan on mobile to open same portal on phone (QRCodeSVG from qrcode.react library)

Step 4 — Customer Decides: Accept or Negotiate

Path A — Direct Confirm (Accept as-is)

Customer clicks "Confirm & Accept";
→ POST /api/negotiations/:quotationId/confirm-portal
→ status: CONFIRMED
→ Socket.io emits 'portal:confirmed' to server
→ Sales team gets notification instantly
→ Portal shows "Deal Confirmed ■" message

Path B — Negotiate (Counter-offer)

Customer clicks "Request Changes / Negotiate";
■■■ Types a message (e.g., "Can you reduce to ■4,00,000?");
■■■ Optionally enters a counter price
■■■ Clicks "Send";
→ POST /api/negotiations/:quotationId/negotiate
→ { message: "...", proposedPrice: 400000 }
→ status: UNDER_NEGOTIATION
→ Socket.io emits 'negotiation:message' to 'portal_{token}' room
→ Sales team sees message in real-time

Step 5 — Live Negotiation Chat

CustomerPortal ← Socket.io → Sales Rep's screen

Customer types message → socket event: 'negotiation:message'
↓
Sales Rep sees it instantly
↓
Sales Rep responds (from main app)
↓
Customer's portal auto-updates (no page refresh)

Socket room joining:

```
// In CustomerPortal.jsx
const socket = io(&#x27;http://localhost:5000&#x27;, { withCredentials: true });
socket.emit(&#x27;join_portal&#x27;, { token });

socket.on(&#x27;negotiation:message&#x27;, (msg) => {
  setMessages(prev => [...prev, msg]);
});
```

Step 6 — After Negotiation Resolved

Sales Rep updates quotation (new price/terms) in main app
→ status reverts to SENT_TO_CUSTOMER with updated values
→ Customer's portal refreshes showing new price
→ Customer reviews updated quotation
→ Customer clicks "Confirm & Accept";
→ status: CONFIRMED → deal done ■

Step 7 — Post-Confirmation (Customer side)

Portal shows:
■■■ "Your order has been confirmed!";
■■■ Reference number
■■■ "You will receive invoice shortly";
■■■ Download button (if PDF available)

Customer Flow Summary



FULL SYSTEM FLOW — All Roles Together

ADMIN sets up:
Products → Discount Tiers → Upsell Rules → Warehouses → Users

SALES REP creates:
Quotation (Draft) → adds products → LiveMarginBar shows risk
→ Submits → PENDING_MANAGER

SALES MANAGER approves (Stage 1):
Reviews discount/risk → Approves or Rejects
If discount > 10% → PENDING_FINANCE

FINANCE approves (Stage 2):
Final check → Approves → APPROVED

ADMIN/REP sends to customer:
Clicks "Send" → portal URL generated → email sent
Status: SENT_TO_CUSTOMER

CUSTOMER opens magic link:
Views quotation → Negotiates or Accepts
Socket.io: real-time chat with sales team

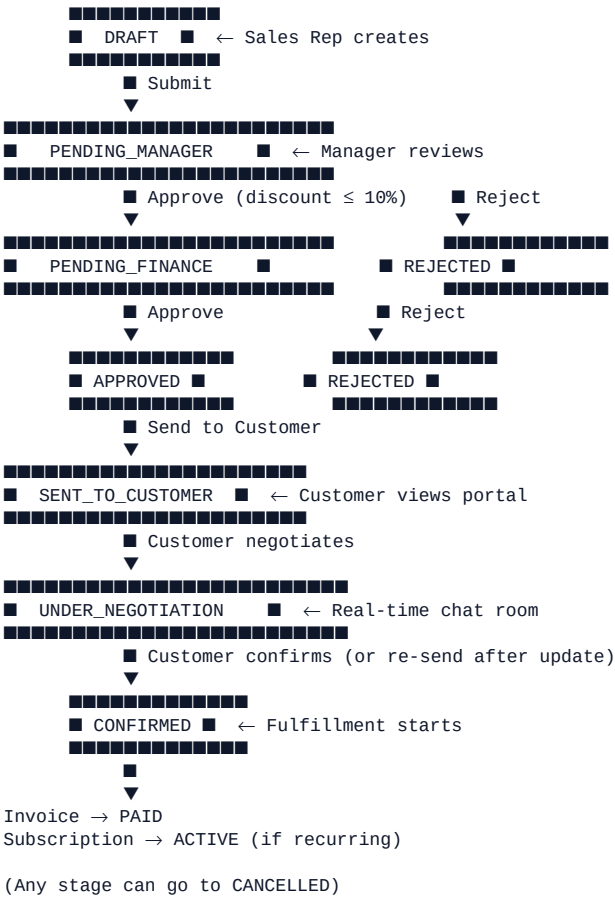
CUSTOMER confirms:
Status: CONFIRMED
Socket.io: instant notification to sales team

ADMIN processes fulfillment:
Checks warehouse stock → split allocation
Accepts split → stock deducted

FINANCE creates invoice:
PDF generated (pdfkit) → email sent → customer pays
Status: PAID

ADMIN activates subscription (if SaaS):
Subscription created → recurring billing tracked
MRR shown on dashboard

Key Data States (Quotation Status Machine)



Role Permission Summary

Action	ADMIN	MANAGER	FINANCE	SALES REP	CUSTOMER
View Dashboard	■	■	■	■	■
Manage Products	■	■	■	■	■
Manage Users	■	■	■	■	■
Configure Discount Tiers	■	■	■	■	■
Create Quotation	■	■	■	■	■
Submit Quotation	■	■	■	■	■
Approve (Stage 1)	■	■	■	■	■
Approve (Stage 2)	■	■	■	■	■
Send to Customer	■	■	■	■	■
View Customer Portal	■	■	■	■	■
Negotiate in Portal	■	■	■	■	■
Confirm Deal	■	■	■	■	■
Process Fulfillment	■	■	■	■	■
Generate Invoice	■	■	■	■	■
Manage Subscriptions	■	■	■	■	■
View Warehouses	■	■	■	■	■